1 Installing Python

First check if you have Python installed by opening up powershell on Windows and type python –version or terminal on MAC/Linux type python3 –version. If you do not get an error, Python is installed and you can move on to the next section.

If you get an error, Python is not installed and I would reccomend the following tutorials in installing Python and understanding its uses

Windows https://www.youtube.com/watch?v=YKSpANU8jPE

Mac https://www.youtube.com/watch?v=VJjeMdl6yWA

2 Installing Necessary Packages

Python packages are pre-built code that anyone can install to make use of in the future. This code requires several packages that you may not have installed on your machine, so the first step is to install the packages.

To do this navigate to where you have the 'film_triple_channel_dosimetry_main_branch' in the terminal.

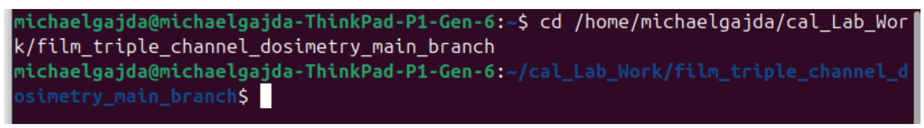

Figure 1: Example use of the cd command to get to the proper directory. Note that in windows '\' is used instead of '/'

This can be done either by using the 'cd' command in the terminal followed by the file path to 'film_triple_channel_dosimetry_main_branch' as above or right clicking while inside the 'film_triple_channel_dosimetry_main_branch' and clicking on the open in terminal option.

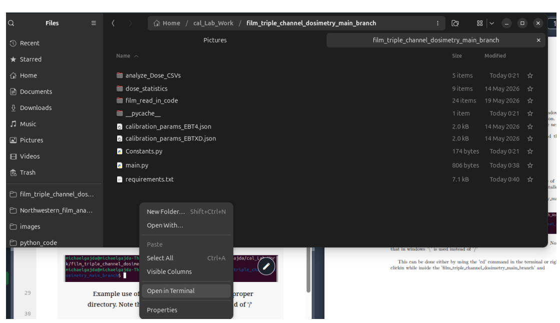

Figure 2: Example of accessing terminal through right clicking in the proper folder

After you are in the proper directory, you are ready to install the necessary packages.

If you want to modify the base version of python copy the following commands into your already open terminal

Windows/Mac/Linux pip install -r requirements.txt

If you run the above command you will never have to refer to this section again

If you instead want to create a virtual environment run the following commands

Mac/Linux python -m venv venv # Create the virtual environment source venv/bin/activate # Activate it pip install -r requirements.txt # Install packages

Windows python -m venv venv # Create the virtual environment venv\Scripts\activate # Activate it pip install -r requirements.txt # Install packages

you are now ready to proceed to run the code in your current terminal window. If you open a terminal window in the future you will have to run the following commands

Mac/Linux source venv/bin/activate

Windows venv\Scripts\activate

These commands will have to run in every new terminal window you decide to open if you want to run the program.

3 Running the Program

3.1 Requirements for Running this program

This program requires both a pre and post scan of the film to run if you do not have one or the other this program will not work.

There are also the python requirments mentioned in the previous section.

3.2 Setting Up Your Directory

For the easiest time I recommend having separate directories for every pre and post scan pair and having them in the folder you have the main code in such as the following example with the 'pre_post_example' folder

Figure 3: An example of the ideal directory setup where 'triple_channel' has the code and 'pre_post_example' contains the pre and post irradiaiton scans

3.3 Navigate to the Appropriate Terminal

Please open a terminal window and navigate to the appropriate directory as mentioned in the Installing Necessary packages section and if you used a virtual environment to install packages please re-enable it now with the following commands

Mac/Linux source venv/bin/activate

Windows venv\Scripts\activate

3.4 Setting Constants

Inside the 'film_triple_channel_dosimetry_main_branch' there exists a file called 'Constants.py' this is the one file that has to be edited in order to have the proper title for plots when opening the file with a txt editor you will see the following.

```
type = 'EBT4'
import os
doseCurveJsonPath = os.path.join(os.path.dirname(__file__), f"calibration_params_{type}.json")
film_labels = [0, 1, 2, 3, 4, 5, 6]
```

Figure 4: View of the inside of the console file

3 things may need to be changed

3.4.1 type

For the type variable either put 'EBTXD' or 'EBT4' depending on the film that you are attempting to analyze.

3.4.2 doseCurveJsonPath

For this variable you do not have to do anything as long as you put EBTXD or EBT4 in all caps for type.

Note that this file does not contain a way to make the .json file used for the calibration so please approach Michael Gajda to make a new .json for new calibration data.

3.4.3 Film Labels

This corresponds to the film id you have assigned to each individual film to enable better tracking.

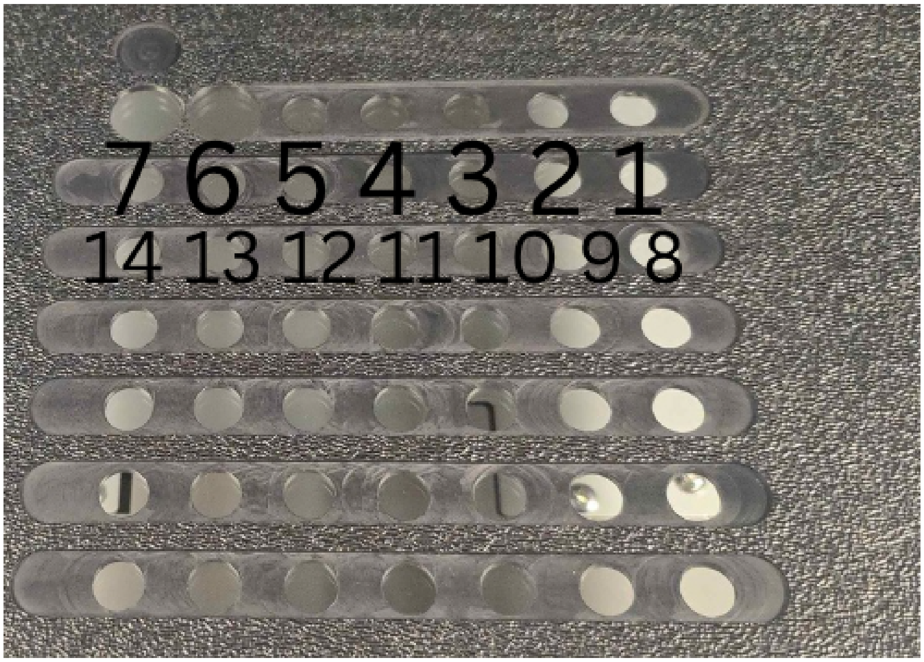

Figure 5: An example mapping of films in the tray

in order to film this in properly you must start with an open bracket '[' as in the example then put your desired number '0' in the case of the example follow by a ',' then continue the pattern till the last number where instead you do not put a comma and finish with ']' . The first number you put will correspond will correspond to the tray slot labeled 1 in the above picture and so on.

3.5 Running File

In the open terminal window with the directory set to 'film_triple_channel_dosimetry_main_branch' run the file with

Windows python .\main.py

Mac/Linux python3 ./main.py

this will start the analysis sequence if the correct constants were entered

3.5.1 Analyzing Films

When first running the file you will be met with this dialogue window

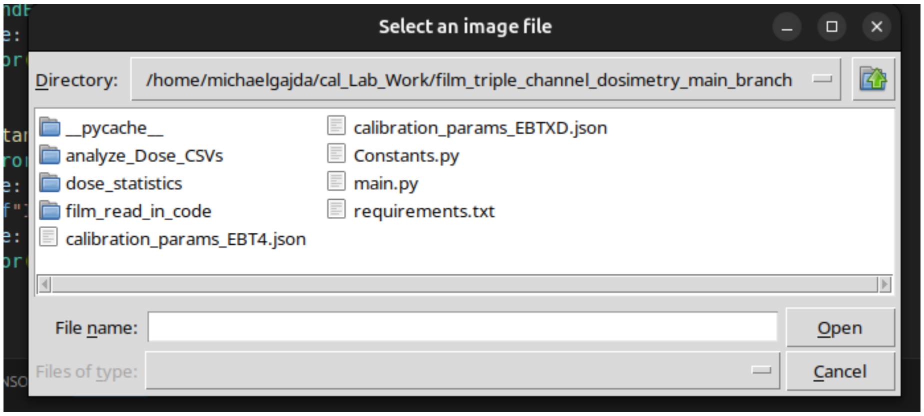

Figure 6: First dialogue window

from here navigate to your pre/background scan and select it

if you have already analyzed this file there will be a dialogue box saying if you want to continue with the previous analysis of the file if you want reanalyze the file type ’n’ and press enter.

the first dialogue after ward will ask you for downscalling size unless you used a dpi higher than 400 this will not matter simply hit enter

afterwards you will be greated by this clustering dialogue box

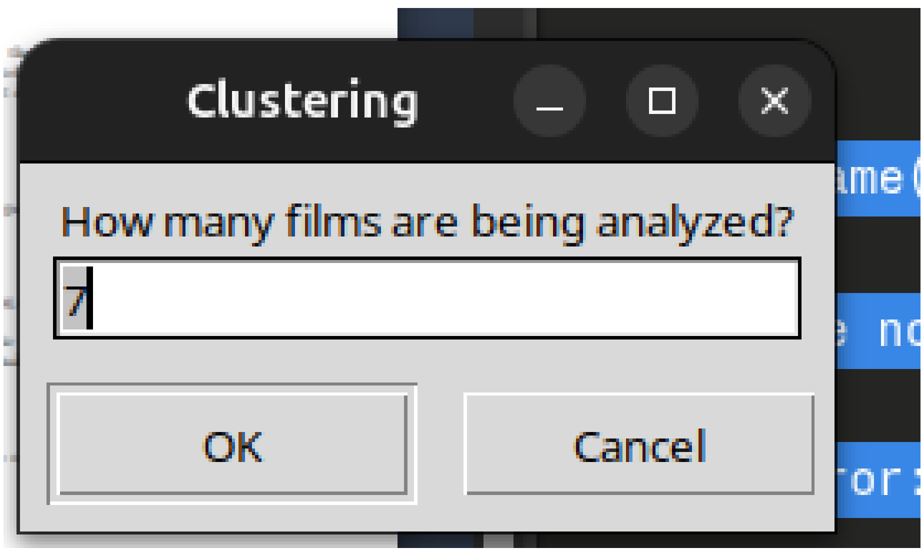

Figure 7: Clustering

here enter the number of films you will be analyzing note that this corresponds to the type you entered in the constants file the rest of the dialogues that pop up after should correspond to the setup used as of 5/21/2026, but in case it changes in the future in order they correspond to

the number of nistCalibration filters used in the scan

the OD values of the nist calibtation values

the number of background shapes to draw in the next step

the threshold for the background

3.5.2 Selecting ROI

After the dialgoue boxes you will be met with the file you selected first select the NIST filters are like shown below by left clicking and dragging to make a rectangle highlighting the region of interest (ROI). After you are satisfied press enter.

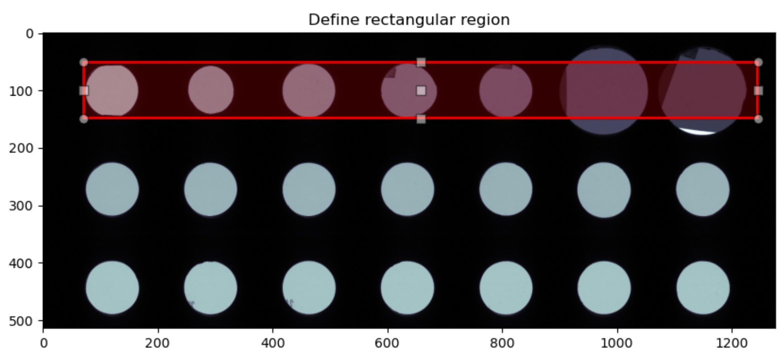

Figure 8: Example of selecting the NIST ROI

afterward you will be shown your selected roi and be asked to select a background region. Select only the tray parts present in the image like so

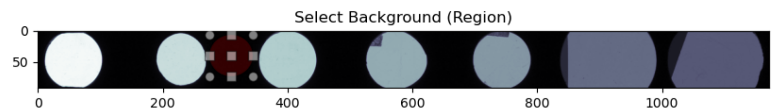

Figure 9: Example background selection

after selecting the background region and pressing enter you will be met with the region of interest division shown below

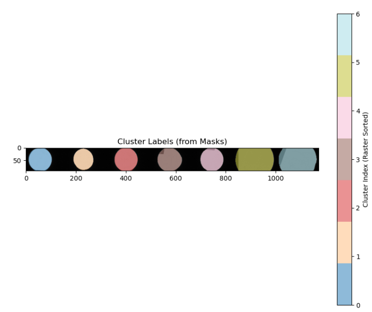

Figure 10: Ideal NIST classification

The image shown should match the above exactly in terms of ordering of the colors. If this is the case exit out of the window and press enter on the next dialogue box to accept the classifcation

There will be occasions where the algorithm will misclassify such as the following

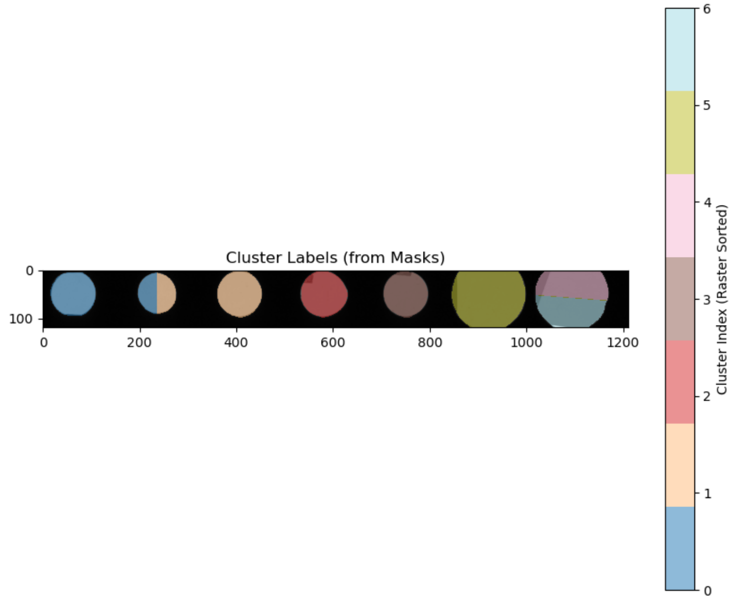

Figure 11: Wrong classification

if this is the case exit out of the window and press 'n' then enter on the next dialogue box to restart the process.

Keep in mind you should have your rectangle be as tall to only cover the smaller hole of the nist filters for the best results as the example shows.

Afterwards you will be met with your selected film again at this time select the ROI containg the films you want to analyze like the below example

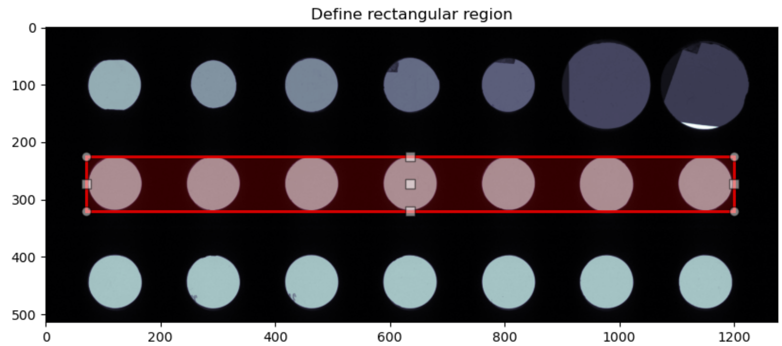

Figure 12: Example selection

afterwards the process you completed for the nist filters will start again. Complete the process and you are offically done with the pre/background scan! You will be shown the NIST cal curves for each channel just to make sure everything is alright and then you will be prompted to select your post file repeat the process and you are on you way to getting the dose histograms!

3.6 After runtime

After the file has finished running you will have the following folders in your original directory containing the pre and post scan

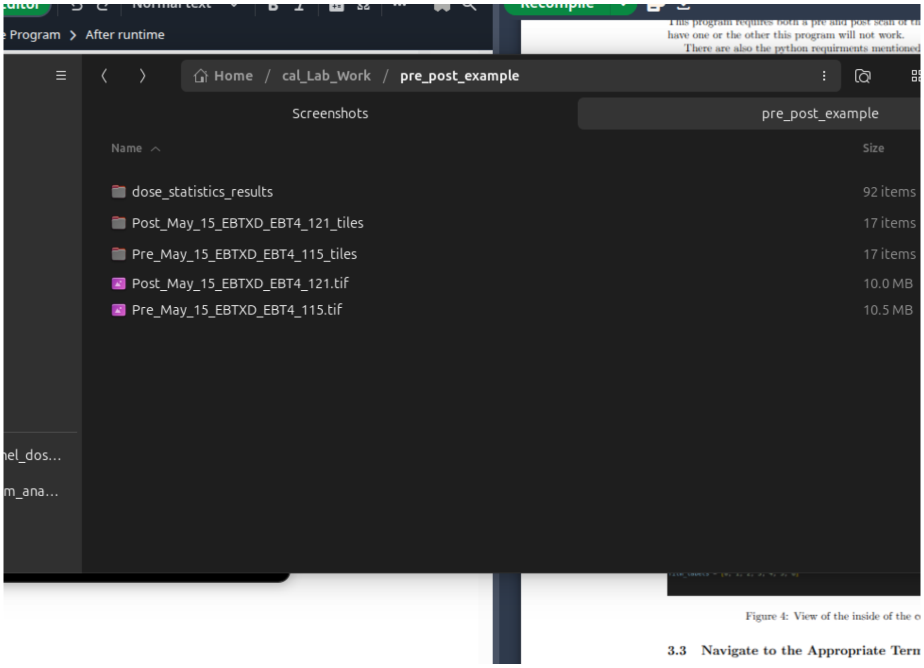

Figure 13: Example results

The folders labeled after your scan followed by '_tiles' contain the sections of film used for the calculation as well as their OD curves in the form of .npy files for each of the three channels.

'dose_statistics_results contains the OD profiles and histograms for each film as well as the individual dose csvs used for each of the films and the histograms folder inside contains the dose histograms for the files.

That is all contantact Michael Gajda if you have any questions.